Clear Claims, Clear Slides, Clear Delivery

Scientific communication in one hour

A talk with audience activity and questions



The same fictional result can inform or mislead, depending on how it is told

Says nothing	Readmission results
Overstates	Calls slash readmissions by a third
Informs	Patients who got a call had fewer readmissions, but the gap narrowed after accounting for risk

Fictional teaching data, not a real study.

One hour, four decisions

Decide	Claim	Show	Deliver
Who is it for, and what is your message?	Does your wording match your evidence?	Does each slide make its point?	Can you say it in time and answer honestly?

Fictional case throughout: nurse calls after discharge, 850 patients, calls not randomised.

Every talk starts with people who must do something afterwards



Work left to right. If you cannot fill a box, you are not ready to make slides.

Teaching recommendation, drawing on Bourne 2007 and the CDC Clear Communication Index (professional guidance).

A topic names an area; a message makes a claim someone could challenge

TOPIC

Telephone follow-up after hospital discharge

MESSAGE

A nurse call is linked to fewer readmissions, but the evidence does not yet justify full roll-out

Three checks for your sentence

Is it a full sentence?

It has a verb and says something happened.

Could someone disagree?

If nobody could, it is still a topic.

Could you say it in a corridor?

It should work without slides.

Fictional teaching data, not a real study. neu-data teaching recommendation.

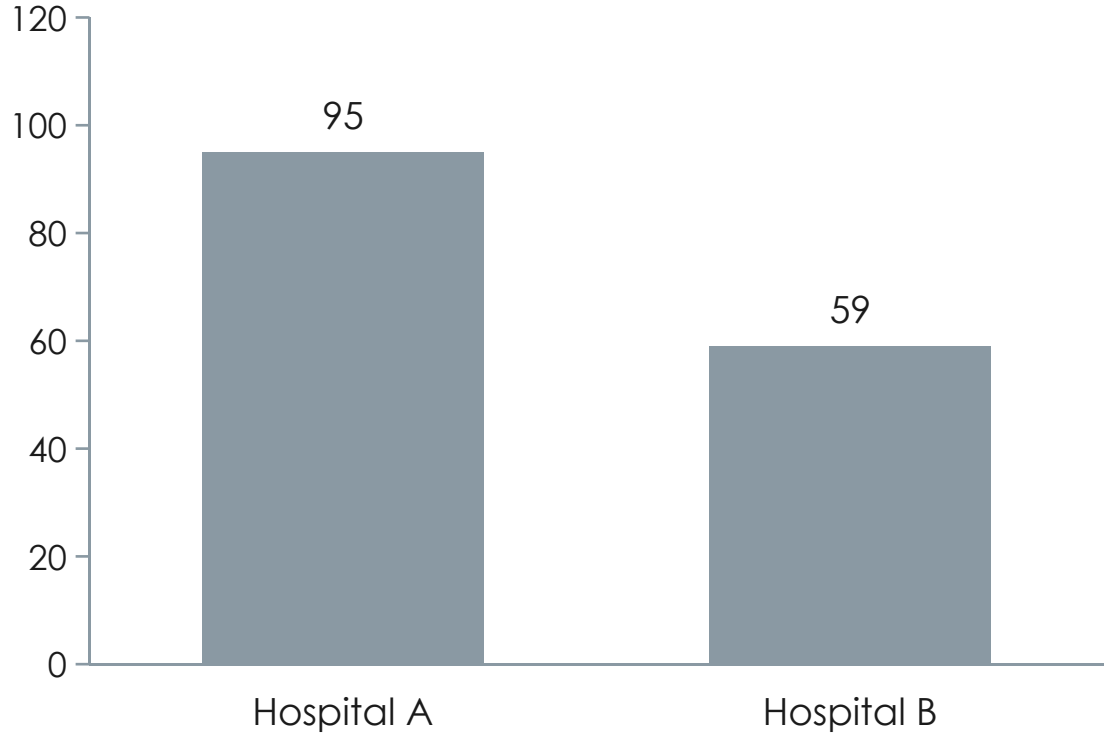
One finding needs a different message in each room

Room	Who they are	What they should do	Your message
Conference audience	Peers in the field, mixed methods knowledge	Consider a new explanation and ask how it applies to their setting	In two hospitals, a nurse call was linked to 6.9 fewer readmissions per 100, but much of that gap reflected who could be reached.
Thesis committee	Examiners who know the field and will probe your choices	Judge that the work is yours, sound and a contribution	I show that reachability confounds observational evaluations of follow-up calls, and I defend the stratified analysis that reveals it.
Hospital management	Decision-makers with budgets, little time, limited methods detail	Decide what to do next	We recommend a six-month staggered trial on two wards before funding full roll-out, because the benefit could be 9.4 fewer to 0.8 more readmissions per 100.

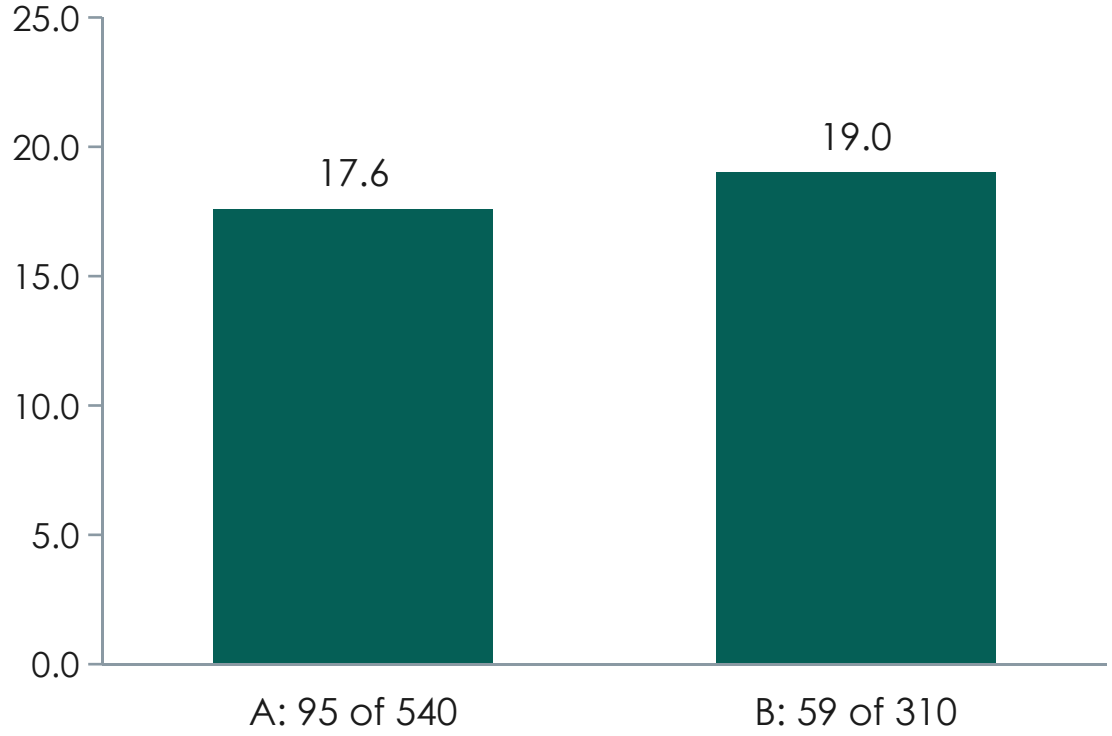
Fictional teaching data, not a real study. neu-data teaching examples.

A count without a denominator can reverse the picture

Counted: Hospital A looks worse



As a rate: Hospital B is higher



Give numerator and denominator every time: 95 of 540, not just 95 and not just 17.6%.

Fictional teaching data, not a real study. Reporting guidance: SAMPL (Lang & Altman 2015).

Report the size of the difference and how uncertain it is

32%

relative reduction alone: sounds large, hides the base

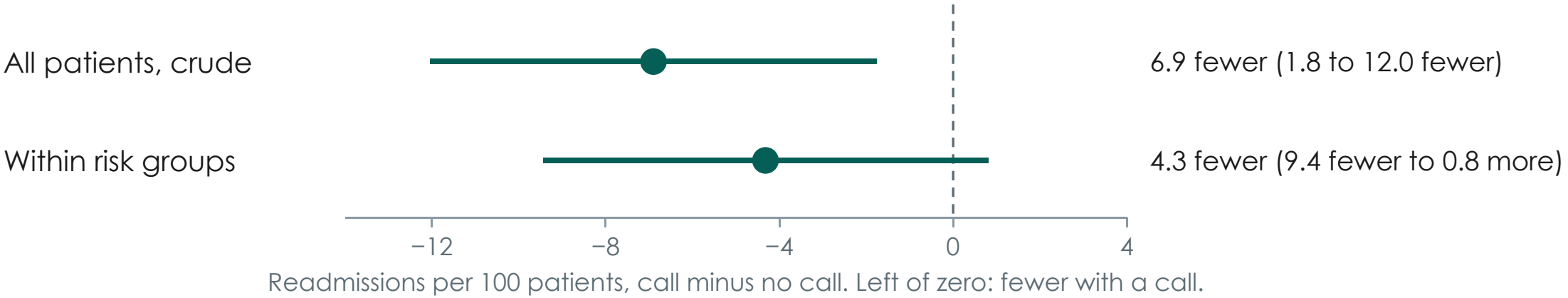
6.9 per 100

absolute difference: 21.5% v 14.6%

1.8 to 12.0

95% confidence interval: the range the data fit

Difference in readmissions



Fictional teaching data, not a real study. Professional guidance: SAMPL, STROBE. Research finding: Akl et al. 2011 (Cochrane), van der Bles et al. 2020.

Choose the verb your design can support

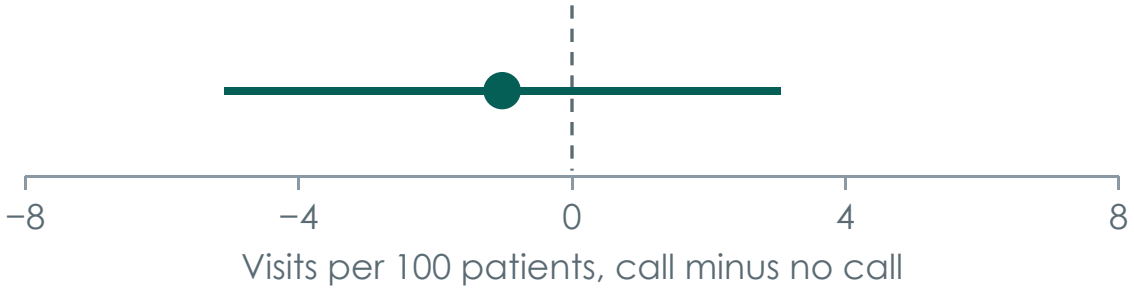
What you have	Verbs that fit	Too strong for it
Observational comparison, as in CareCall	was associated with, was higher among	led to, reduced, prevented
Comparison with adjustment	remained lower after stratifying by risk	the effect of, independently caused
Well-conducted randomised comparison	reduced, in this population	proves, works everywhere
Any single study, in the conclusion	may, suggests, is consistent with	shows conclusively, will improve

The word "association" is not a shield. Many readers still hear a causal claim, so state the design and the main alternative explanation as well.

Research findings: Haber 2018, 2022. Guidance: STROBE, GRADE 26 (for reviews). Wording ladder: neu-data teaching recommendation.

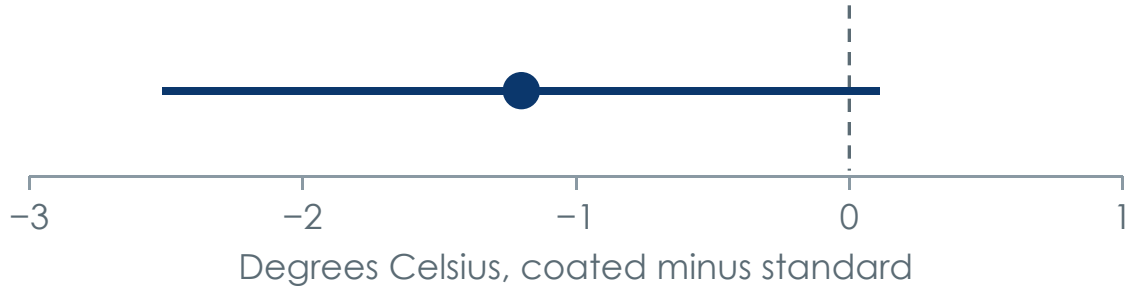
A non-significant result is rarely proof of no effect

CareCall: emergency visits



1.0 fewer per 100, but anything from 5.1 fewer to 3.0 more fits the data.

CoolRoom: classroom temperature



1.2 °C cooler, with 2.5 °C cooler to 0.1 °C warmer fitting the data.

Do not write: "The calls had no effect."

Write: "The study cannot tell us whether calls change emergency visits. The interval includes a useful benefit and a small harm."

Fictional teaching data, not a real study. Review: Altman & Bland 1995, Greenland et al. 2016, ASA statement 2016.

Your turn: fix these three headlines

Activity, 5 minutes. Alone (2), then with a neighbour (3).

"Readmission results"	Topic label. Says nothing has been found.
"Calls slash readmissions by a third"	Causal, and uses the relative figure alone.
"Calls had no effect on emergency visits"	Treats an uncertain result as proof of none.

Facts: readmissions 60 of 412 (14.6%) with a call, 94 of 438 (21.5%) without. Not randomised. Emergency visits 9.7% and 10.7%, difference 1.0 fewer per 100 (5.1 fewer to 3.0 more).

Fictional teaching data, not a real study.

A good headline is a sentence, gives the size and keeps the caution

Instead of "Readmission results"

Patients with a call were readmitted less often: 14.6% v 21.5% (not randomised)

Instead of "Calls slash readmissions by a third"

Readmissions were 6.9 points lower with a call, and 4.3 points lower within risk groups

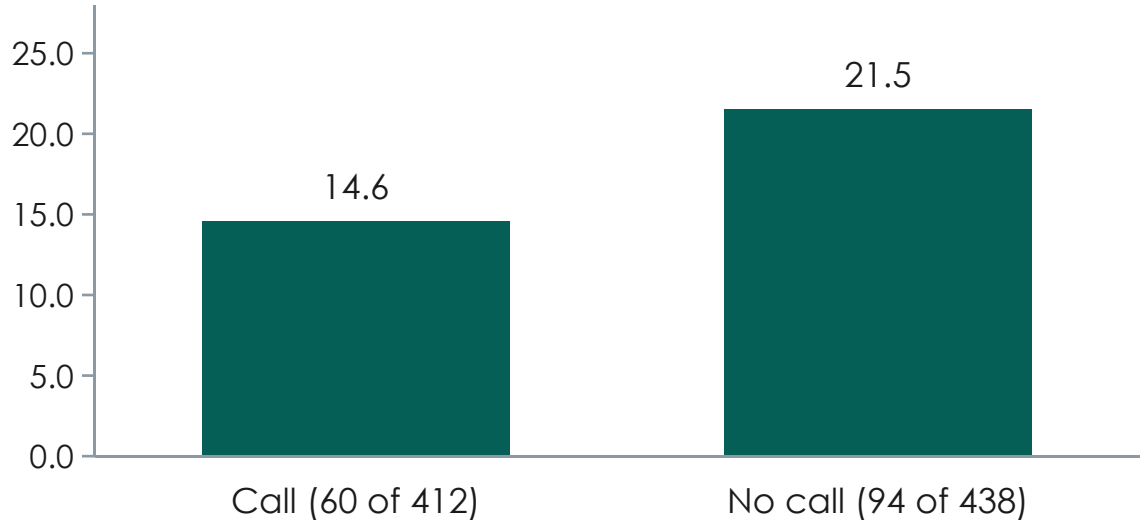
Instead of "Calls had no effect on emergency visits"

Emergency visits were similar, but the data cannot rule out a useful benefit

Fictional teaching data, not a real study. Model answers: neu-data. Other good versions exist.

A slide headline states the claim; the body shows the evidence

Readmissions were lower among patients who received a call



Not randomised. Fictional data.

Headline

A full sentence that makes the claim.

Evidence

A chart, diagram, photo or table that supports it.

Spoken detail

You explain. Secondary detail goes in your notes.

What the evidence shows

In one experiment this bundle beat topic headlines with bullets. It has critics, and no study tests each part alone. Treat it as a strong default.

Research finding: Garner & Alley 2013 (a bundle of changes, recorded talk, students). Critique: Farkas 2010. Method: Alley, Penn State.

Slide-count, word-count and font-size rules are rules of thumb, not findings

Rule you may hear	What sources actually say	Use it like this
One slide per minute	Bourne: more than one visual a minute is too many, as his rule of thumb. Naegle: a minute or less. Other tested talks ran faster.	A planning check. A dense slide needs longer; a build needs less.
Six words, six lines	No primary source found. Advice varies. The direction (less text while you speak) has reasoned support.	Ask what must be read, and cut the rest to your notes.
Minimum 24 or 30 point	Published advice ranges from 18 to 32 pt. Legibility depends on screen size and distance.	Stand at the back of the actual room, or test at reduced size.

Our search found no experiment setting these numbers. Sources in the notes. Guidance labelled as such.

BEFORE

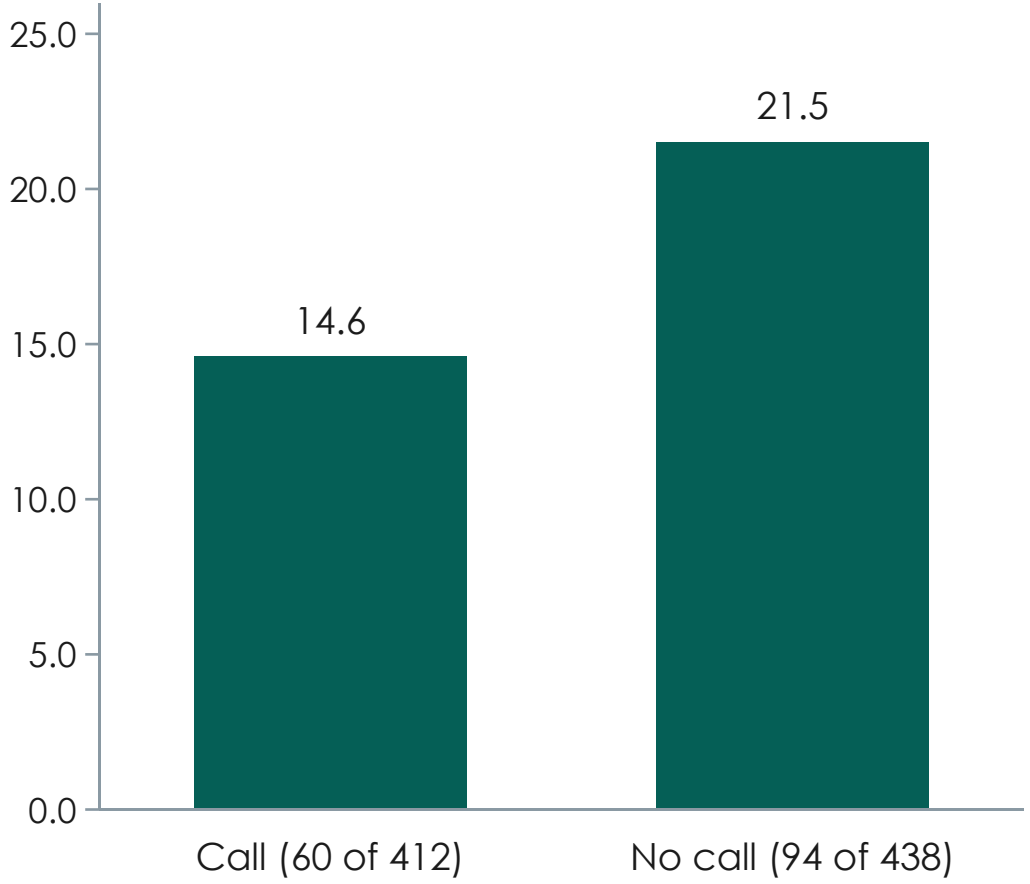
Results

- Total sample size was 850 patients discharged from medical wards at Hospitals A and B during the study period
- 412 patients received a nurse telephone call and 438 patients did not receive a nurse telephone call
- In the call group 60 patients (14.6%) had an unplanned readmission within 30 days
- In the no-call group 94 patients (21.5%) had an unplanned readmission within 30 days
- The risk difference was 6.9 percentage points (95% CI 1.8 to 12.0)
- Patients at high discharge risk made up 29% of the call group and 45% of the no-call group
- After stratification by discharge risk the difference was 4.3 points (9.4 lower to 0.8 higher)
- This suggests that the intervention may have had a beneficial impact on readmission outcomes

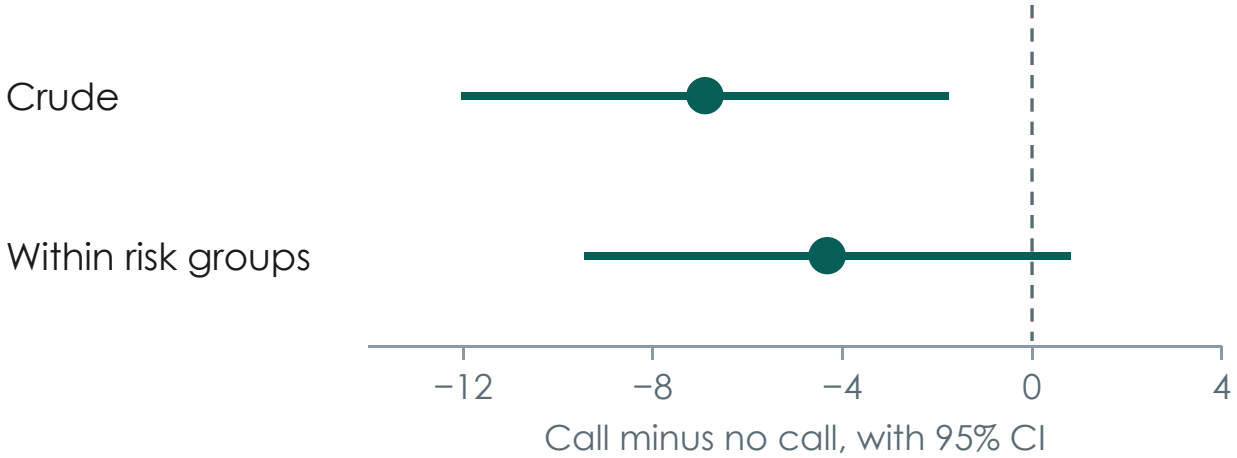
Fictional teaching data, not a real study.

AFTER

Patients who got a call had fewer readmissions, but the gap narrowed after accounting for risk



Difference in readmissions per 100 patients



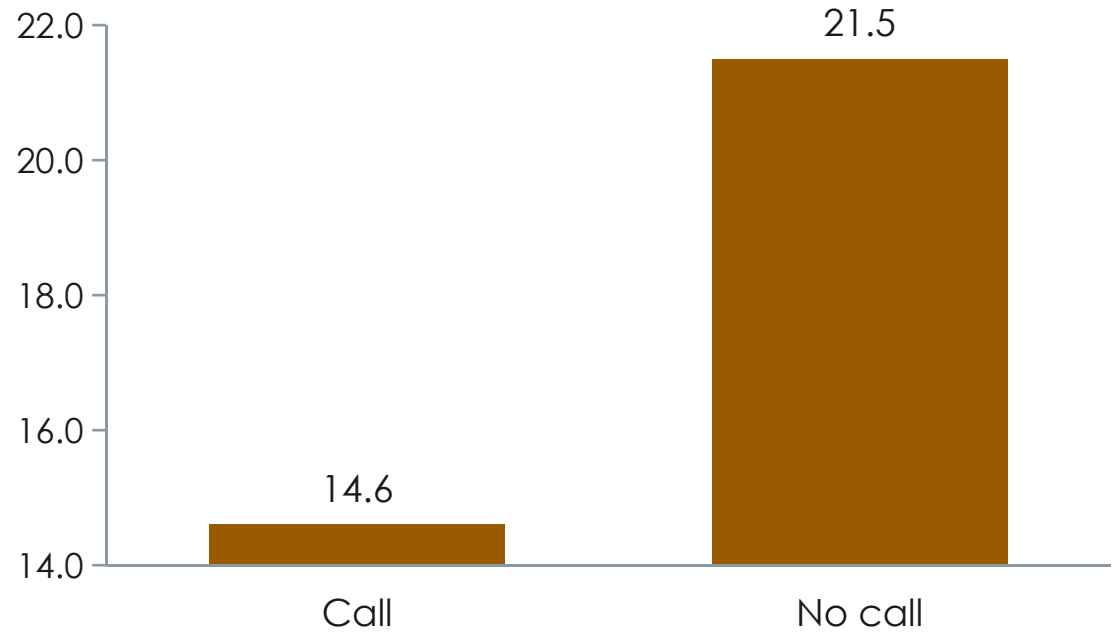
High-risk patients were harder to reach: 29% of those called, 45% of those not called.

Calls were not randomised.

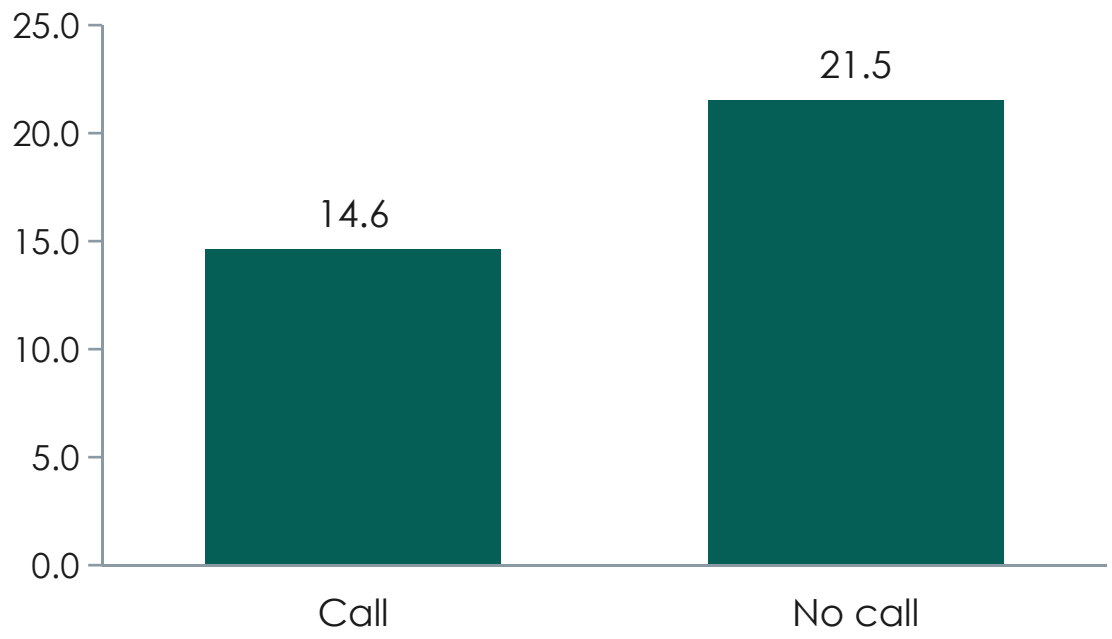
Fictional teaching data, not a real study.

The same numbers can look dramatic or modest, so choose honestly

Axis starts at 14: the gap looks huge



Axis starts at 0: the gap is in proportion

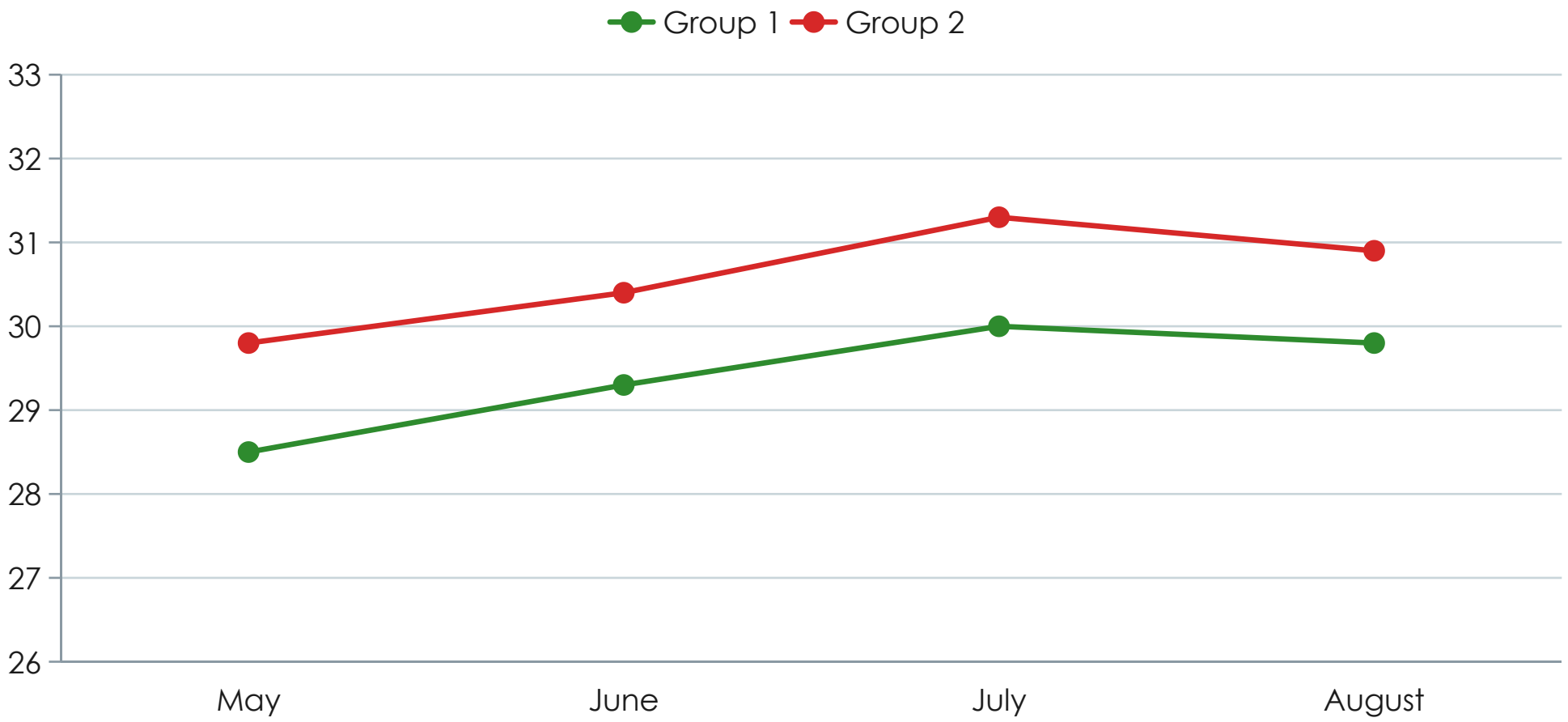


Bars encode length, so bars start at zero. Dots and lines encode position and may use a narrower range, if the range is labelled and the change is meaningful.

Fictional teaching data, not a real study. Research findings: Pandey et al. 2015, Correll et al. 2020 (online samples). Guidance: Wilke 2019.

BEFORE

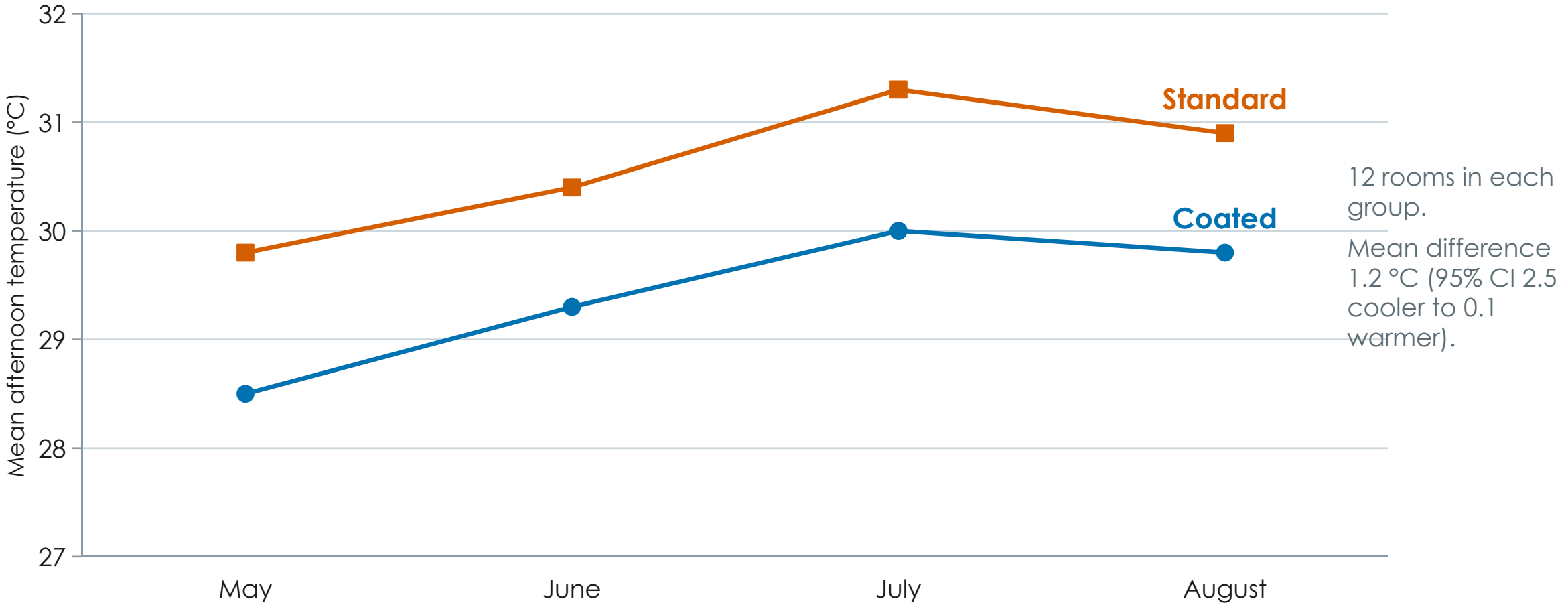
Temperature comparison by month



Fictional teaching data, not a real study.

AFTER

Coated classrooms averaged 1.2 °C cooler, but the interval includes no difference

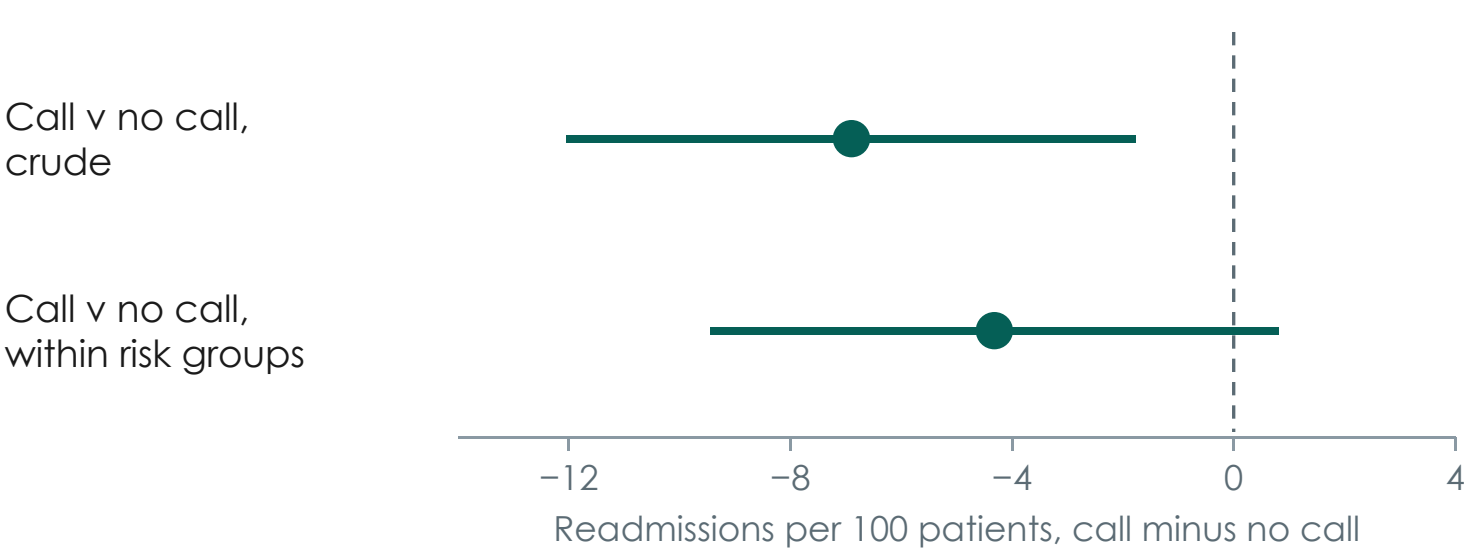


12 rooms in each group.

Mean difference 1.2 °C (95% CI 2.5 cooler to 0.1 warmer).

Fictional teaching data, not a real study.

Every data figure must answer five questions for the audience



Adults discharged from medical wards at two hospitals, n = 850. Bars show 95% CI.
Source: fictional CareCall data.

1. **Axes**
   What is on each axis?
2. **Units**
   Per 100? Degrees? Days?
3. **Denominators**
   Out of how many?
4. **Uncertainty**
   What do the bars mean?
5. **Source**
   Where did the data come from?

Fictional teaching data, not a real study. Teaching checklist: neu-data, drawing on SAMPL and Rougier et al. 2014.

Explain every figure in four moves

1 What it shows	"Each dot is the difference in readmissions, per hundred patients."
2 Where to look	"Look at the top line, and the dashed line at zero."
3 What you see	"With a call there were about 7 fewer. The interval stays below zero."
4 Why it matters	"So there is a signal. The next line shows how much survives comparing like with like."

Fictional teaching data, not a real study. Teaching recommendation, drawing on W3C Making Events Accessible and the assertion-evidence tutorial.

Rehearse against the real clock and decide in advance what to cut

Say it aloud →	Time it →	Cut →	Run for a listener
Full run, standing, with the clicker. Silent reading does not count.	Use the real limit. Note the time at each section.	Remove before you speed up.	Use the feedback prompts. Then revise.

Mark your slides before you rehearse

Must keep	Should keep	Can cut
Question, main result, limit, conclusion	Second result, method detail	Extra examples, background already known

If you run over on the day, jump to your conclusion. Never skip it.

Professional advice: Bourne 2007, Naegle 2021, Penn State. We found no study of rehearsal timing itself.

Listen, repeat, answer briefly, then check

Listen to the end → **Repeat or rephrase** → **Answer briefly** → **Check it landed**

Question type	What to do	You could say
Critical	Find the valid part and answer that	"That is a fair concern. Reachability may explain part of this."
Unclear	Check what they mean	"Do you mean the crude result or the stratified one?"
Out of scope	Say so and offer a route	"We did not study cost per patient. I can follow up afterwards."
Unknown	Say so plainly	"I do not know. I will check and reply by email."

Professional and teaching guidance: Streeter & Miller 2011, Dunphy 2007, NIU, W3C Making Events Accessible.

AI can help you edit and rehearse; you verify every fact, reference and claim

Useful for

Tightening sentences you wrote

Finding hard-to-say phrases

Suggesting likely questions

Timing a script

Check by hand

Every number against your table

Every reference: open it

Every verb: did the claim grow?

Every summary against the source

Do not

Paste patient or confidential data into tools your organisation has not approved

List AI as an author

Skip disclosure when required

In one 2025 study, most language models tested widened scientific claims when summarising, and asking for accuracy did not help. A polished sentence can still be an overstated one.

Research findings: Walters & Wilder 2023, Peters & Chin-Yee 2025. Policy: ICMJE 2026, COPE. Check your journal and institution.

Your questions

What would you like to ask?

If it would help to start somewhere:

A slide your supervisor insists on

A result that is not significant

How many slides you really need

Before your next talk: one sentence, one honest headline, one timed run

One sentence

Your message, with its caution.

One honest headline

On every slide, no stronger than the data.

One timed run

Aloud, with a listener, against the limit.

References and the three-hour workshop materials are in the course repository.



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